

Arghya Bhattacharjee

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LinkedIn: linkedin.com/in/arghya-bhattacharjee876 | GitHub: github.com/Arghya876 | Portfolio:

<https://arghya-bhattacharjee.vercel.app/>

Profile Summary

Software developer and information technology student with a strong foundation in **software development, artificial intelligence, machine learning, and IoT**. Skilled in Java, Python, JavaScript, React.js, Node.js, MongoDB, and AIML, with hands-on experience in developing web applications and intelligent systems. Experienced in building real-time IoT solutions, AI-powered applications, and secure web platforms through academic, internship, and patent-backed projects. Passionate about leveraging technology to solve real-world problems and continuously expanding expertise in software engineering and emerging technologies.

Education

Degree	Institution	Board/University	Year
B.Tech (IT) CGPA: 8.1	JIS College of Engineering	JIS College of Engineering	2026
Higher Secondary (12th) 84.40%	Aranghata Upendra Memorial Institution (H.S.)	WBCHSE	2022

Skills:

Languages: Java, Python, JavaScript, HTML, CSS

Concepts: REST API, OOPs, DSA, JWT

Frameworks & Libraries: React.js, Node.js, Express.js

Databases: MongoDB, MySQL, SQL

Tools & Platforms: Git, GitHub, Arduino IDE, Postman

Special Interests: Artificial Intelligence, Machine Learning, IoT

Experience

PRODIGY INFOTECH

Aug 2024 – Sep 2024 | WEB DEVELOPMENT INTERN | Mumbai, Maharashtra, India

- Developed dynamic web applications using HTML, CSS, JavaScript, and APIs.
- Integrated OpenWeatherMap API for real-time weather data.
- Enhanced UI/UX for improved user experience across devices.

Projects:

NovaCart: MERN Stack E-Commerce Platform

- Full-Stack Multi-Vendor E-Commerce Platform.
- Developed customer, seller, and admin dashboards with role-based access control.
- Implemented secure authentication and authorization using JWT and protected routes.
- Built RESTful APIs for product management, order tracking, cart operations, and user management.
- Designed responsive and modern user interfaces optimized for desktop and mobile devices.
- Integrated MongoDB for managing products, users, orders, and analytics data efficiently.
- Focused on scalability, performance optimization, and secure API communication.
- Tech Stack:** React.js, Node.js, Express.js, MongoDB, JWT, Vite, Tailwind CSS.
- Link:** <https://novacart-arghya876.vercel.app/>
- GitHub:** <https://github.com/Arghya876/Novacart>

Little Guardian: Smart Baby Cradle and Safety System Using IoT and ML (Final Year Project)

- IoT-Based Monitoring Platform.
- Built backend APIs to process and store real-time IoT data using Firebase and MongoDB.

- Designed scalable data models for feeding logs, alerts, and monitoring analytics.
- Implemented secure endpoints with Spring Security, ensuring protected access to sensitive data.
- Focused on system reliability and performance while handling asynchronous sensor inputs.
- **Tech Stack:** Java, Spring Boot, MongoDB, Firebase, IoT.
- **Link:** <https://little-guardian-frontend.onrender.com/>
- **GitHub:** <https://github.com/Arghya876/Little-Guardian>

AI-Based LPG Gas Leakage Detection & Fire Alert System (*Patent Holder*)

Jan 2024 – Apr 2024

- Developed an AI-integrated IoT system to detect LPG gas leaks and fire hazards in real time.
- Used gas sensors and flame sensors connected to a microcontroller for detection.
- Incorporated AI algorithms for early hazard prediction and prevention.
- Automated WhatsApp notifications to alert homeowners and emergency contacts.
- **Tech Stack:** Arduino, Python, AI/ML models, IoT sensors, and Twilio API.
- **Link:** <https://github.com/Arghya876/AI-based-LPG-Gas-Leakage-Detection-Fire-Alert-System>

Google reCAPTCHA Security Form using Django

Oct 2024

- Integrated Google reCAPTCHA to prevent automated sign-ups.
- **Tech Stack:** Python, Django, Google reCAPTCHA API.
- **Link:** github.com/Arghya876/Google-reCAPTCHA-Security-Form-using-Django

Face Emotion Recognition

Nov 2024

- AI-powered facial emotion detection using deep learning models.
- **Tech Stack:** Python, OpenCV, TensorFlow/Keras, NumPy.
- **Link:** github.com/Arghya876/Face-Emotion-Recognition

Certifications

- NPTEL Project Management (8 Weeks) – IIT Roorkee (Oct 2024 – Dec 2024)
- Smart Irrigation System (IoT)
- Python Programming Essentials Bootcamp (3 Days)
- JavaScript Essentials 1
- Google Cybersecurity Professional Certificate